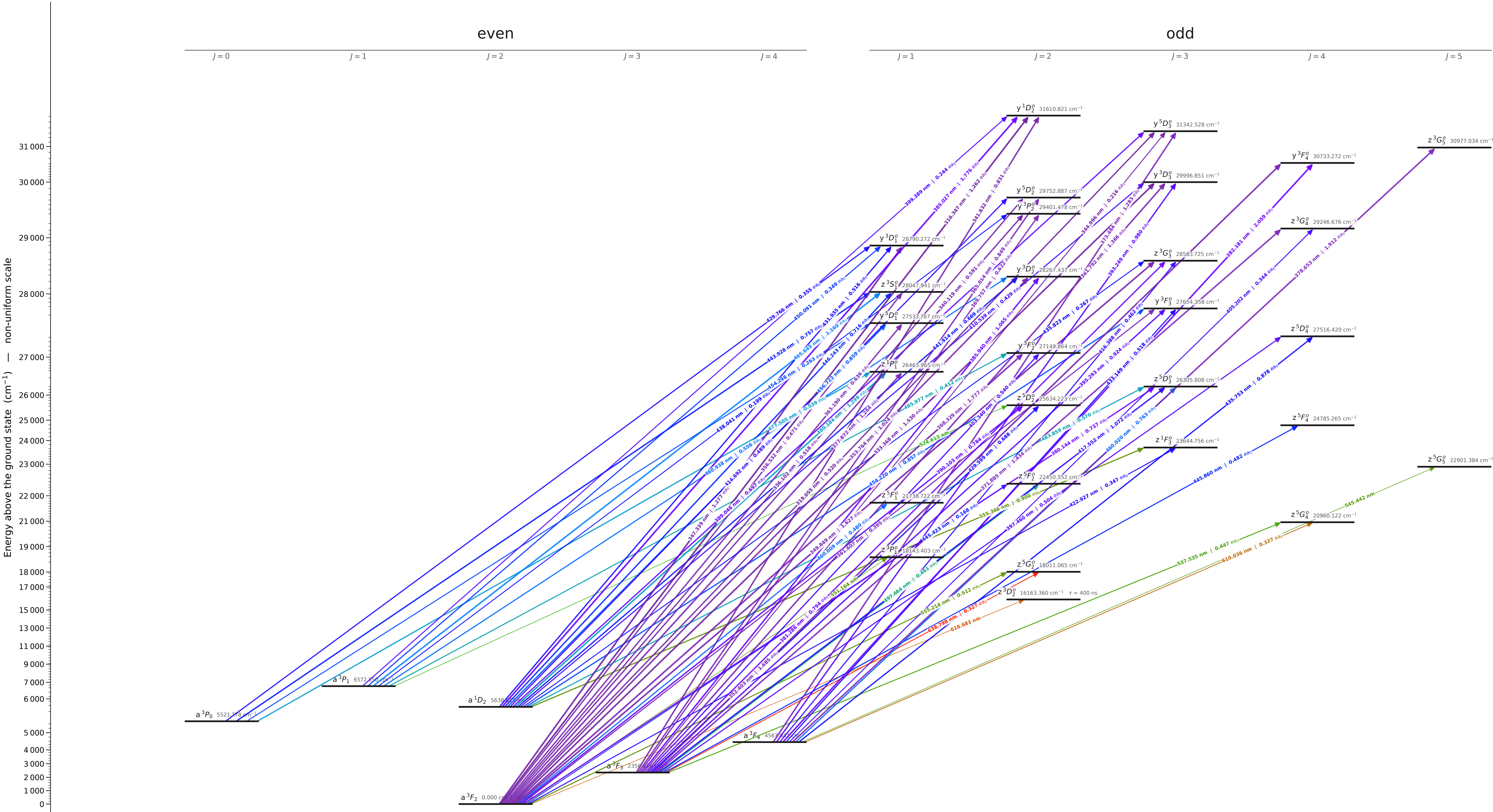
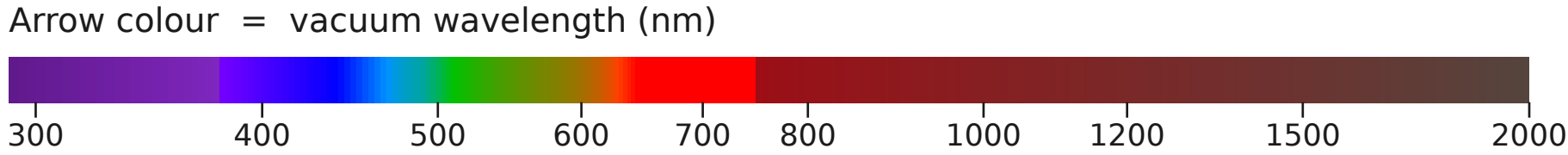


Hafnium

Hf (Hf I). Strongest classified transitions below 2 μm (NIST): 34 levels, 74 lines.

Hf⁺ ionisation limit 55047.90 cm⁻¹ = 6.82507 eV (181.660 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||J\rangle\rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ³ F ₄ – z ⁵ G ₃ ^o	545.4424	545.2908	549.6318	–	–	–	–	545.29 nm line: reference line for the change of nuclear...
a ³ F ₂ – z ³ P ₁ ^o	551.1645	551.0114	543.9256	–	–	–	–	551.01 nm line: isotope-shift and hyperfine laser spectro...
a ³ F ₂ – z ⁵ G ₃ ^o	555.2143	555.0601	539.9581	0.512	6.200e+05	–	–	555.06 nm line: isotope-shift and hyperfine laser spectro...
a ³ F ₂ – z ³ D ₂ ^o	618.6832	618.5121	484.5654	–	–	0.3979	–	618.5 nm ground-state line used for optical isotope-shift...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
a ³ F ₂ – z ³ P ₁ ^o	551.164	180-178	378.5(15)
a ³ F ₂ – z ⁵ G ₃ ^o	555.214	180-178	-960.1(15)

Shift = ν(first isotope) – ν(second isotope).

¹⁷⁹Hf hyperfine structure (I = 9/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
a ³ F ₂	-71.430	706.000	F=5/2: +1109.31 F=7/2: +550.43 F=9/2: +8.37 F=11/2: -384.49 F=13/2: -466.37
a ³ F ₃	-50.810	931.000	F=3/2: +1265.07 F=5/2: +944.09 F=7/2: +549.02 F=9/2: +134.18 F=11/2: -230.62 F=13/2: -460.03 F=15/2: -453.19
a ³ F ₄	-43.460	1619.000	F=1/2: +1698.16 F=3/2: +1481.19 F=5/2: +1143.66 F=7/2: +721.72 F=9/2: +265.95 F=11/2: -158.59 F=13/2: -472.40 F=15/2: -581.52 F=17/2: -377.53
a ¹ D ₂	-47.680	-905.000	F=5/2: +109.69 F=7/2: +338.75 F=9/2: +407.00 F=11/2: +144.76 F=13/2: -655.37
z ³ D ₂ ^o	-46.930	740.000	F=5/2: +855.40 F=7/2: +367.39 F=9/2: -75.04 F=11/2: -333.16 F=13/2: -237.37
z ⁵ G ₃ ^o	128.740	2802.000	F=5/2: -131.89 F=7/2: -907.17 F=9/2: -1203.47 F=11/2: -495.40 F=13/2: +1859.16

Measured A, B constants; sources in the data file.